

Received 25 April 2025, accepted 21 May 2025, date of publication 2 June 2025, date of current version 9 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3575203

RESEARCH ARTICLE

Attention-Based Transfer Learning for Efficient Obstructive Sleep Apnea (OSA) Classification on Snore Sound

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This work was supported by the Petchra Pra Jom Klao Ph.D. Research Scholarship from the King Mongkut's University of Technology Thonburi (Scholarship No. 44/2564).

ABSTRACT Polysomnography (PSG) is currently the gold-standard technique for classifying sleep apnea disorders. Yet, it is costly and requires an expert to score the severity, making it impractical for self-screening and home use. Snore sound classification with Deep Learning (DL) is a promising approach and has gained increasing interest due to its relationship with abnormal breathing conditions in both time and frequency domains. This study proposes an attention-based transfer learning model for non-invasive detection of obstructive sleep apnea (OSA) using audio signals. Mel-spectrograms and MFCC features were input into the MobileNetV3-Large to extract deep features. A modified SENet was implemented to provide suitable channel attention for the extracted features. The attention-based features are then classified into normal and abnormal snore events. The study evaluates model performance using 10-fold cross-validation on sound data of the 70 adult OSA patients of an open-source PSG-Audio dataset. Results show that utilizing both Mel-Spectrogram and MFCCs as input features significantly enhances classification performance compared to single-feature models. In addition, the MobileNetV3-Large with modified SENet significantly outperforms the combination of other pre-trained and attention mechanisms. Specifically, the proposed model provides an accuracy of $92.576 \pm 0.910\%$, a sensitivity of $92.906 \pm 1.928\%$, a specificity of $92.269 \pm 2.740\%$, a precision of $92.173 \pm 3.024\%$, and an F1-score of $92.486 \pm 1.326\%$ on the binary classification. Its performance also shows statistically significant improvement when benchmarked with other existing OSA classification models. Our proposed model demonstrates suitable potential for portable device-based sleep apnea monitoring applications.

INDEX TERMS Obstructive sleep apnea (OSA), snore sound, deep learning (DL), OSA classification, transfer learning.

I. INTRODUCTION

Obstructive sleep apnea (OSA) is the most common type of sleep-related breathing disorder. It is characterized by involuntary stops in breathing from repeated upper airway obstruction during sleep, which is associated with arousals with or without oxygen desaturations. It comprises up to one-seventh of the world's adult population of age 30-69 years, or approximately one billion people worldwide. Out of them, 425 million people suffer from moderate to severe

OSA, defined by an Apnea-Hypopnea Index (AHI) ≥ 15 events per hour [2]. OSA is a chronic condition that can affect both physically and mentally adverse health conditions. It is also a risk factor for the further development of serious diseases such as hypertension, diabetes mellitus, coronary artery disease, heart attack, myocardial infarction, ischemic stroke, depression [3], [4] etc. Polysomnography (PSG) is currently the gold-standard technique widely used to screen sleep apnea patients. It accurately identifies sleep disorders by monitoring the physiological changes of body signals using multiple sensors [4], [5], [6]. The technique requires a specialized laboratory equipped with numerous sensors and

The associate editor coordinating the review of this manuscript and approving it for publication was Ghulam Muhammad¹.

devices that must be attached to a patient. The process might affect a patient's sleep quality. Despite its high accuracy in screening for sleep disorders, the technique is expensive and requires an expert to score the severity, so it hinders self-screening applications for home use.

This limitation makes it necessary to explore a simple, efficient, and non-contact way to diagnose sleep disorders. Recently, research on disease detection using DL has been actively conducted in Computer Aided Diagnosis (CAD) and achieved good performance in various medical fields. Snore sound analysis with DL is also one of the promising approaches [7], [8]. It has gained increasing interest due to its relationship with abnormal breathing conditions and the distinct differences in the distribution of acoustic features between simple snore sounds, and OSA snore sounds in both time and frequency domains [9], [10], [11], [12], [13].

The DL approach to OSA classification based on snore sounds have proven efficient in several studies over recent years [14], [15], [16], [17]. Shen et al. [14] used Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) to distinguish between normal and OSA snore sounds, employing Mel-frequency Cepstral Coefficients (MFCC), Linear Predictive Coding Cepstral Coefficients (LPCC), and their combination (LPMFCC) features. Their results showed that the LSTM with MFCC inputs achieved the highest accuracy of 87% in binary classification and could estimate the AHI, which is crucial for determining OSA severity. Cheng et al. [15] used LSTM with multi-inputs to detect respiratory event-related snore sounds. MFCC, Mel Filter Banks (Fbanks), Short-time Energy, and Linear Prediction Coefficients (LPC) were combined based on multi-LSTM branches for classifications. Their model achieved 95.3% accuracy for binary classification on snore sounds of normal individuals against those abnormal sounds of OSA, and 81.6% for three-class classification on snore sounds of normal individuals against those normal and abnormal sounds of OSA. Recently, Ding et al. [16] proposed a hybrid model combining a pre-trained VGG19 and LSTM to classify normal and apnea-hypopnea snore sounds and estimate AHI from the Mel-spectrograms. Their model achieved 85.21% accuracy, despite challenges of imbalanced data, highlighting the need for further improvements in detection accuracy. Although these methods yielded satisfactory classification performance, their network architectures necessitate substantial computational resources, hence restricting their practical applicability, especially in portable or resource-constrained applications. Another limitation in previous studies is the absence of attention mechanisms to direct the model's focus toward the most informative segments of an audio signal, thereby enhancing both classification accuracy and interpretability. Furthermore, it is noteworthy that these deep learning methods typically employ distinct internal datasets for their research, rendering direct performance comparison challenging.

Fortunately, more open-source snore sound datasets have recently become available for researchers to study. The

PSG-Audio dataset from the Sleep Study Unit of the Sismanoglio-Amalia Fleming General Hospital of Athens [1] is among the open-source PSG data publicly available. This dataset is potentially an important benchmark for evaluating and comparing classifiers for OSA, particularly those based on audio recordings. The PSG-Audio dataset contains snore sound recordings covering all OSA severity types and annotations of nasal and respiratory events by medical experts. One challenge of this dataset is the number of overlaps between the duration of normal and apnea/hypopnea snore events, which has never been reported in other studies. The snore sounds were recorded from two types of microphones, i.e. contact (tracheal) microphones and non-contact (ambient) microphones, which affects OSA classification performance [18], [19]. A tracheal microphone is in direct contact with the patient's trachea and is less noise sensitive. Its sound quality generally is better than that of an ambient microphone placed in the bedroom [18]. However, the ambient microphone is more comfortable for patients, though its effectiveness can be influenced by the distance from the patient's head [19].

Upon recording the snore sound, features are extracted as the input for a DL model. The Mel-spectrogram is one of the most widely used features for sound classification because it provides a time-frequency representation that captures both temporal and spectral characteristics [13]. Mel-spectrograms enable sound classification to be approached as an image classification problem by visualizing audio signals as an image, leveraging image processing techniques and models. CNN has proven effective in extracting the most informative features from images and is considered the current state-of-the-art technique for image classification.

The size of the training dataset significantly impacts classification performance in deep learning-based OSA diagnosis. However, building sufficient labeled datasets is challenging due to the high cost and effort for acquiring ground truth labels. Transfer learning addresses these limitations by exploiting a pre-trained model on a large, general dataset, and fine-tuning to a specific, smaller task [20], [21]. This technique leverages the knowledge acquired from the initial task to improve performance and stability and reduces training time for new related tasks. Conceptually, the knowledge used to transfer across domains is encoded into the learned feature representation. With the new feature representation, the performance of the target task is expected to improve significantly. Transfer learning is particularly useful for disease diagnosis in fields with limited datasets like medical image classification [22], [23], [24] and OSA classification.

Currently, several pre-trained models based on CNN, such as AlexNet [25], VGGNet [26], ResNet [27], MobileNet [28], and DenseNet [29], have been proposed. AlexNet [25] pioneered the field of deep learning, making it useful for understanding fundamental principles. VGGNet [26] is well-suited for transfer learning but is resource intensive. ResNet [27] known for its deep networks and residual connections, excels in complex tasks requiring deep feature extraction. MobileNet [28] is optimized for efficiency, low

resource usage, and speed, making it ideal for mobile and real-time applications. However, it sacrifices some accuracy compared to more complex models. On the other hand, DenseNet [29] is stronger in accuracy and feature reuse due to its dense connections, but it is computationally heavier and less suited for mobile or embedded devices. Each model's unique strengths and weaknesses suit different use cases and computational environments. For OSA screening applications, MobileNet offers a balance of efficiency and lower computational resource requirements.

The MobileNetV3 model [30] is a member of the MobileNet family of efficient CNNs, designed to achieve an optimal balance between accuracy and efficiency. Specifically, this makes it a powerful choice for AI applications on resource-constrained environments such as mobile and embedded platforms. MobileNetV3 is available in two variants, i.e. MobileNetV3-Large and MobileNetV3-Small, to suit different performance and computational requirements.

MobileNetV3-Large consists of 21 layers of depth wise separable convolutions, neural architecture search (NAS), squeeze-and-excitation (SE) modules, hard swish (H-Swish) activation, and inverted residuals. It is designed for tasks requiring precise predictions, such as high-resolution image classification and complex object detection. The pre-trained MobileNetV3-Large model was chosen as the base model due to its moderate size and high accuracy. This model effectively offers a good balance between performance and efficiency, especially in scenarios requiring low computational resources and highly suitable for future applications on portable devices.

Traditional CNNs struggle to detect correlations with distant pixels due to their focus on spatially adjacent pixels. Recent advancements incorporate attention mechanisms to address this limitation. Attention mechanisms enhance performance by focusing on the most informative parts of the data. Integrating these mechanisms into convolution blocks has shown great potential, often complementing traditional CNN structures. The Squeeze-and-Excitation Network (SENet) [31] is an attention mechanism to enhance feature recalibration through channel-wise interdependencies and global information incorporation with minimal additional computation. SENet can be easily integrated into architectures like ResNet and MobileNet for various tasks, including image classification and object detection. It is a valuable tool for enhancing CNN performance and boosting the network's ability to focus on critical information.

This research investigates the possibility of DL approaches to classify OSA using snore sound recordings from a subset of patients from the open-source dataset [1]. Using Mel-spectrograms and MFCCs as sound feature inputs, the proposed model combines a pre-trained MobileNetV3-Large model with a modified attention-based SENet that replaces the global average pooling technique with a flattening technique to extract high-level features. The model finally classifies audio features into normal and abnormal

(apnea or hypopnea) sounds for binary OSA classification. The performance of the proposed model was compared to the existing methods.

Our contributions are presented by expanding upon previous findings in two key areas:

- Presenting an attention-based transfer learning architecture that retains more comprehensive spatial and frequency information, thereby facilitating the extraction of higher-level features essential for accurate classification, and outperforms existing OSA classification models based on 10-fold cross-validation on the same open-source dataset.
- Providing an ablation study and possible alternative combination analysis of the proposed model to prove its advantages.

The remaining sections of the paper are organized as follows: Section II describes the dataset and the proposed method; Section III presents the analysis of the results; and Section IV provides the final conclusions.

II. MATERIALS AND METHODS

The overall process of this study is shown in the block diagram in Figure 1. Snore sounds with labels were obtained from the open-source database. Mel-spectrogram and MFCC features were extracted from the labeled snore segments as inputs for the model. Each feature was input for different branches of the attention-based transfer learning model to classify sound segments into normal sound and abnormal snore sound events. The model training and evaluation were conducted using a 10-fold cross-validation technique.

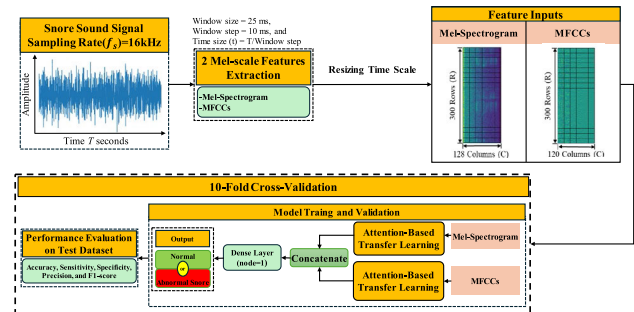


FIGURE 1. The block diagram of the proposed framework.

A. SNORE DATASET

The snore sound data of 70 adult OSA patients (52 male and 18 female) was selected from the open-source PSG-Audio dataset [1], which includes nasal and respiratory event annotations after the automatic rejection of the false-positive apneas. The snore sounds were simultaneously recorded using both high-quality tracheal and ambient microphones. The tracheal microphone was in contact with the patient's trachea, while the ambient microphone was positioned 1 meter above the patient's bedhead. Both microphones were set to record snore sounds at a sampling rate of 48 kHz. For this study, only snore sounds recorded from the ambient

microphone were used, and they were resampled to 16 kHz, a conventional sampling rate widely used in many studies [14], [15], [32].

From the provided annotations, 29,146 snoring and breathing-related sound segments were extracted with an average duration of 17.041 ± 18.058 seconds, as detailed in Table 1. These segments were categorized into two main sound classes: normal sounds and abnormal snore sounds. Normal sounds comprised of normal snores and non-snore sounds. Non-snore sounds were evenly segmented of 18 seconds from non-respiratory events. Abnormal snore sounds consisted of all types of apnea and hypopnea snore sounds, following the American Academy of Sleep Medicine (AASM) criteria for calculating the Apnea-Hypopnea Index (AHI). Therefore, there are 14,578 normal sounds (7,578 normal snore sounds and 7,000 non-snore sounds) and 14,568 abnormal snore sounds, achieving approximately balanced sound classes. Additionally, overlapping segments between abnormal and normal snore events were observed, with an average overlapping duration of 9.595 ± 31.001 seconds in this dataset.

TABLE 1. The distribution of abnormal snore and normal sound segments.

Group of Subjects	Abnormal Snore Sounds				Normal Sounds		Overlap
	Hypopnea	Obstructive	Central	Mixed	Snore	Non-Snore	
52 Males	2,653	7,116	419	1,235	4,981	4,897	2,360
18 Females	1,183	1,714	55	193	2,597	2,103	868
Total	3,836	8,830	474	1,428	7,578	7,000	3,228
	14,568				14,578		

B. FEATURE EXTRACTION

1) MEL-SPECTROGRAM

A Mel-spectrogram is a 2D representation of an audio signal's energy across time and frequency, using the Mel scale to align with human auditory perception [33]. This provides a more accurate representation of pitch differences perceived by the human ear, which is helpful in speech recognition, music analysis, and audio classification based on their frequency analysis, and audio classification based on their frequency contents. Creating a Mel-spectrogram involves several steps. First, a pre-emphasis filter is applied to the audio signal to amplify higher frequencies to balance the frequency spectrum. The audio signal is then divided into small overlapping frames to capture the temporal aspects. Each frame undergoes windowing, typically with a Hamming window, to reduce edge effects. The windowing helps reduce signal discontinuities at the start and end of each frame. In this study, a window size of 25 milliseconds with a 40% overlap was

used, i.e. a step size of 10 milliseconds. Each frame is then Fourier transformed to produce the power spectrum. The power spectrum is passed through a Mel filter bank, which maps the frequencies to the Mel scale, simulating human auditory perception. The number of Mel bands was set to 128. Finally, the logarithm of the filter bank energies is taken from the Mel-spectrogram, representing human's ear logarithmic perception of loudness (sound's energy) over time and frequency on the Mel scale. The formula to convert a frequency f (in Hz) to the Mel scale is:

$$Mel(f) = 2595 \times \log\left(\frac{f}{700} + 1\right) \quad (1)$$

As illustrated in upper part of Figure 1, Mel-spectrograms of each snore sound with different duration times (T) were extracted with a Mel-band size of 128 using the Librosa library [34]. Then, the Mel-spectrograms were resized to a constant size of 300×128 .

2) MEL-FREQUENCY CEPSTRAL COEFFICIENTS

Mel-Frequency Cepstral Coefficients (MFCCs) is among a feature widely used in audio processing [17]. The process of extracting MFCCs is similar to that of Mel-spectrograms, as both involve applying Mel-scaled filters and logarithmic compression. However, MFCC extraction further applies a Discrete Cosine Transform (DCT) to reduce the data into a set of coefficients that capture the most essential features of the sound. In contrast, Mel-spectrograms retain the full 2D time-frequency representation without further transformation. For the MFCCs extraction, the window size of each frame and overlap were set equal to the Mel-spectrogram feature and the number of cepstrum was set to 40. The standard MFCC parameter primarily captures the static characteristics of sound parameters. However, the dynamic characteristics of sound can be better represented by analyzing the spectrum difference of these static features. Thus, the first and second-order differences of the MFCC coefficients were also incorporated. Similar to the Mel-spectrogram extraction process, 120 MFCCs including their first and second-order differences were extracted with the python_speech_features library [35] and then resized to a constant size of 300×120 as shown in the upper part of Figure 1.

C. DEEP LEARNING CLASSIFICATION MODEL

Figure 2 illustrates the models initially designed for single-feature OSA classification. The models have identical structures except for the modified inputs of MobileNetV3-Large to support Mel-spectrogram and MFCCs, respectively. The model structure concatenates a pre-trained MobileNetV3-Large model with the modified SENet model. It is noted that the input features are simply stacked for 3 layers corresponding to the required input dimension of the MobileNetV3-Large model. Fine-tuning the pre-trained MobileNetV3-Large model serves allows the model to extract distinguishable deep features from the snore sound features. These deep features are then utilized by our modified SENet

before proceeding with global average pooling layer and a final classifier consisting of a single-node dense layer with a sigmoid activation function to classify snore sounds as normal sound or abnormal (OSA) snore sounds.

The modified SENet in Figure 3 replaces the global average pooling layer with a flattening layer. The flattening operation converts multi-dimensional feature maps into a 1D representation. This modification preserves all feature map information rather than reducing it to a single average value per channel. Two fully connected layer with the reduction ratio (r) of 16 are followed the flattening layer to produce channel weights. Finally, the channel weights are combined with the inputs based on channel-wise multiplication to enhance the important features.

Figure 4 illustrates our model extension from single-feature input to support two-feature OSA classification. The outputs of our single-feature models are concatenated, and a dense layer with the sigmoid activation function is used for the binary classification as normal or abnormal (OSA) sounds.

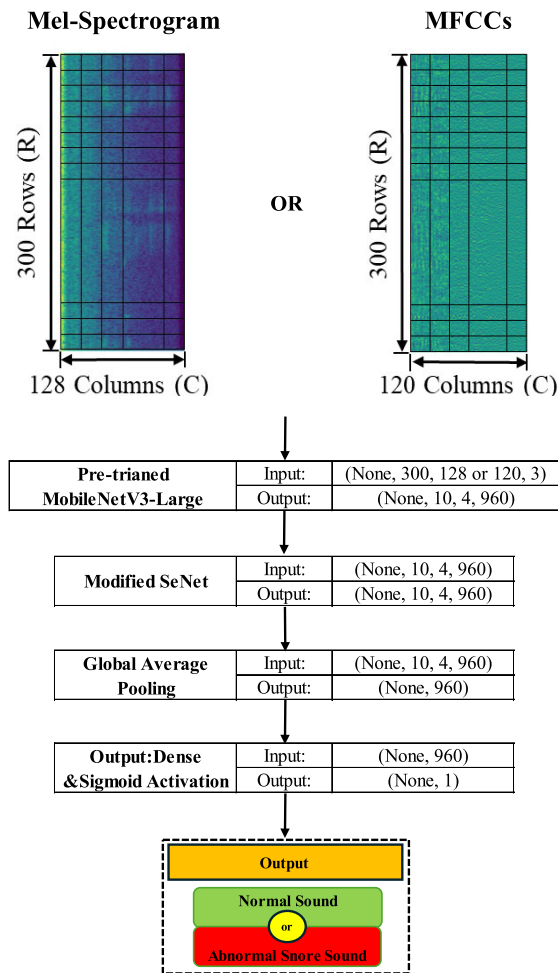


FIGURE 2. The attention-based transfer learning structure for a single feature input, i.e. Mel-spectrogram and MFCCs.

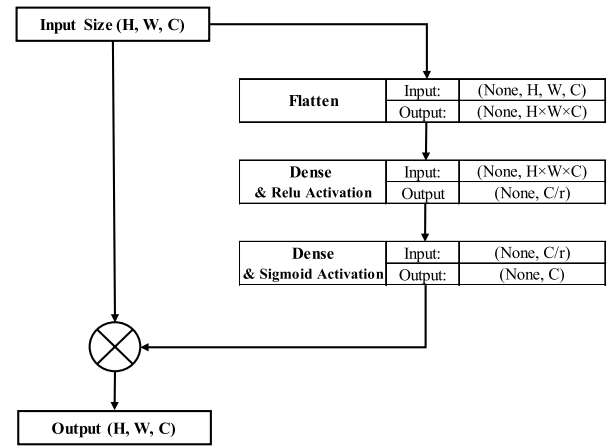


FIGURE 3. The modified SENet structure.

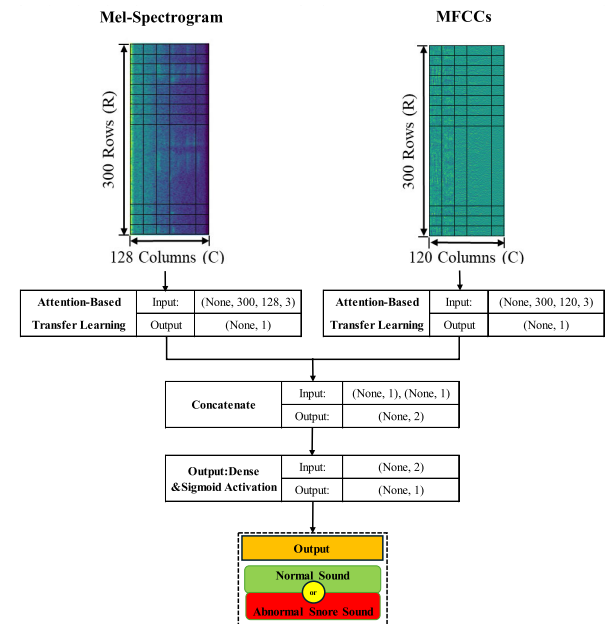


FIGURE 4. The combination of the two transfer learning models for multi-feature inputs on the OSA classification.

D. MODEL TRAINING AND 10-FOLD CROSS VALIDATION

The study employed the 10-fold cross-validation to assess the performance of the proposed model. The sound segments from 70 adult OSA patients were randomly divided into 10 folds, each fold containing segments from 7 different patients. The distribution of sound segments for each class within each fold is summarized in Table 2, while OSA patient distribution details for each fold are provided in Table A1 of Appendix A1. The data in each fold is completely independent of one another. This ensured that data from the same patient did not appear in different folds. On each training iteration, one-fold was designated as the test set, its consecutive fold as the internal validation set, and the remaining 8 folds as the training set. This process was repeated 10 times, hence each fold serving as the test set once and ensuring

a comprehensive and robust model evaluation. Finally, the performance metrics obtained from each fold were averaged to estimate the model's performance. Figure 5 illustrates the 10-fold cross-validation used for the model assessment.

With a large number of training inputs in each cross-validation iteration, the batch size and the number of epochs were set to 128 and 100, respectively. The learning rate was set to 0.001 for training and validation. The adaptive moment estimation (Adam) optimizer was chosen to update network weights. Binary cross-entropy was used as the loss function for binary classification. The model's accuracy and loss were tracked on both the training and validation sets.

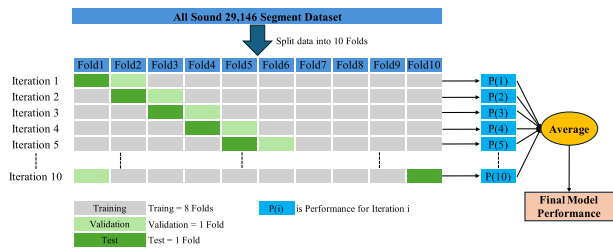


FIGURE 5. A 10-fold cross-validation. 8 training, 1 internal validation, and 1 test folds are cyclically permuted.

TABLE 2. The number of abnormal snore and normal sound segments within each fold.

Fold Number	Number of Sound Segments		
	Abnormal Snore Sounds	Normal Sounds	Total
Fold1	1,026	1,862	2,888
Fold2	1,359	1,572	2,931
Fold3	1,308	1,490	2,798
Fold4	1,515	1,434	2,949
Fold5	1,509	1,412	2,921
Fold6	1,578	1,375	2,953
Fold7	1,640	1,307	2,947
Fold8	1,608	1,319	2,927
Fold9	1,452	1,460	2,912
Fold10	1,573	1,347	2,920
Total	14,568	14,578	29,146

E. CLASSIFICATION EVALUATION

The model performances were evaluated by comparing the confusion matrix between actual and predicted values on the test dataset for each iteration. The classification performance was assessed in terms of average accuracy, sensitivity, specificity, precision, and F1-score.

Accuracy indicates how close the classification results are to the actual values. Sensitivity, also known as the true positive rate, refers to the proportion of actual positives that are correctly identified. Specificity, known as the true negative rate, represents the proportion of actual negatives that are correctly identified. Precision, or positive predictive value, measures the proportion of positive predictions that are correct. The F1-score, the harmonic mean between sensitivity and precision, provides a balanced measure that considers both false positives and false negatives. These performances were calculated as:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \times 100\% \quad (2)$$

$$Sensitivity = TP / (TP + FN) \times 100\% \quad (3)$$

$$Specificity = TN / (TN + FP) \times 100\% \quad (4)$$

$$Precision = TP / (TP + FP) \times 100\% \quad (5)$$

$$F1 = 2 \times Precision \times Sensitivity / (Precision + Sensitivity) \quad (6)$$

where TP represents abnormal snore sounds that are correctly classified, TN represents normal sounds that are correctly classified, FP represents normal sounds that are mistakenly classified as abnormal, and FN represents abnormal snore sounds that are mistakenly classified as normal.

III. RESULTS

The study's results are divided into four sub-sections, each addressing a distinct aspect of model evaluation: (1) input feature selection, which is a critical role in enhancing model learning performance; (2) comparison of pre-trained CNNs, which evaluates the performance of various pre-trained models to determine the most suitable one for combination with the modified SENet; (3) ablation study, which examines the impact and significance of the modified SENet model; and (4) comparison with benchmark models, which assesses the proposed approach with benchmark models based on the same data.

In all studies, 10-fold cross-validation is used for performance evaluation on each model. The average performances are summarized in terms of accuracy, sensitivity, specificity, precision, and F1-score. In addition, a paired t-test with a two-tailed hypothesis using p-value = 0.05 was conducted to validate the statistical significance of the improvements achieved by the proposed method.

A. FEATURE SELECTION

The classification performance of the proposed MobileNetV3-Large model with the modified SENet using different input sound features, i.e. Mel-Spectrogram, MFCCs, both, is summarized in Table 3.

The results show that utilizing a single Mel-Spectrogram feature yields relatively high classification performance with an accuracy of $90.952 \pm 1.184\%$, a sensitivity of $91.315 \pm 3.180\%$, a specificity of $90.801 \pm 3.006\%$, a precision of $90.646 \pm 3.648\%$, and an F1-score of $90.870 \pm 1.470\%$.

TABLE 3. The average classification performances of the proposed model with different input features.

Input Feature	Average Classification Performance (mean $\pm$ standard deviation)				
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-score (%)
Mel-Spectrogram	90.952 $\pm$ 1.184*	91.315 $\pm$ 3.180*	90.801 $\pm$ 3.006*	90.646 $\pm$ 3.648*	90.870 $\pm$ 1.470*
MFCCs	85.767 $\pm$ 1.614*	83.915 $\pm$ 3.727*	87.508 $\pm$ 2.434*	86.732 $\pm$ 4.031*	85.220 $\pm$ 2.932*
Mel-Spectrogram and MFCCs	92.576$\pm$0.910	92.906$\pm$1.928	92.269$\pm$2.740	92.173$\pm$3.024	92.486$\pm$1.326

*= indicates the difference than the best performer (bold) with statistical significance (p-value < 0.05)

These results suggest that the Mel-Spectrogram is highly effective in feature extraction for OSA classification. On the other hand, when using only MFCCs as the input feature, the model's performance clearly declines across all evaluation metrics. This suggests that MFCCs alone are less effective in extracting the necessary classification features for accurate predictions.

However, when both Mel-Spectrogram and MFCCs are utilized, the classification is improved with statistical significance across all evaluation metrics. It yields the highest accuracy of 92.576 $\pm$ 0.910%, a sensitivity of 92.906 $\pm$ 1.928%, a specificity of 92.269 $\pm$ 2.740%, a precision of 92.173 $\pm$ 3.024%, and an F1-score 92.486 $\pm$ 1.326%. These results confirm that utilizing both features enhances the model's ability to learn more information from the feature input.

B. PRE-TRAINED CNNs COMPARISON

To assess choices of different pre-trained CNNs when combined with the modified SENet for OSA classification performance, we evaluated MobileNetV3-Large alongside several other models, i.e. VGG19, InceptionV3, XceptionNet, and EfficientNetB0, as shown in Table 4.

Among all the models, the MobileNetV3-Large with the modified SENet outperforms the others with statistical significance across all performance metrics. While using VGG19 as a based model clearly shows strong performance better than EfficientNetB0, XceptionNet, and InceptionV3, it is less than the proposed model. EfficientNetB0 performs slightly better than XceptionNet, while InceptionV3 achieved the lowest performance.

These results confirm that the combination of MobileNetV3-Large and the modified SENet consistently achieves superior performance compared to other configurations on all metrics, reinforcing MobileNetV3-Large as the most effective base model for OSA classification task.

C. ABLATION STUDY

To further validate the effectiveness of our proposed attention mechanism was evaluated with and without the traditional SENet model. In addition, it was also evaluated on another attention mechanism, i.e. an Efficient Channel Attention

(ECANet). The performance comparison with the proposed model in Table 5 provide insights into impact of different attention mechanisms on classification performance and further validate the effectiveness of our modified SENet approach in combination with MobileNetV3-Large.

The results show that using SENet with MobileNetV3 Large improves the classification performance compared to using MobileNetV3-Large alone with an accuracy of 90.538 $\pm$ 0.954%, a sensitivity of 89.926 $\pm$ 2.136%, a specificity of 91.427 $\pm$ 3.330%, a precision of 90.994 $\pm$ 4.228%, and an F1 score of 90.354 $\pm$ 1.432%, showing that attention mechanisms can enhance feature extraction from the snore sound.

Among different attention mechanisms, the proposed model achieves the highest performance, with an accuracy of 92.576 $\pm$ 0.910%, a sensitivity of 92.906 $\pm$ 1.928%, and an F1-score 92.486 $\pm$ 1.326%. Although its specificity and precision may not show statistically significant improvements, they remain higher than those of other models, showing its effectiveness as the most reliable approach for OSA classification.

D. PERFORMANCE BENCHMARKING

Table 6 presents a comparison between the proposed model and benchmark models for OSA classification. Several DL models from previous studies [14], [15], [16] were selected due to their strong performance in OSA classification using snore sounds, as reported in the literature, and are considered widely recognized benchmarks. All models were evaluated and compared using the same open-source PSG-Audio dataset [1]. The models from these previous works were recreated following their provided details and tested under the same conditions.

The results highlight the statistically significant performance of our proposed model, which clearly outperforms other benchmark models. Among all benchmark models, the four-branch LSTM introduced by Cheng et al. [15], utilizing four input features, i.e. MFCCs, Fbanks, Short time Energy, and LPC, yield the highest performance with an accuracy of 89.171 $\pm$ 1.989%, a sensitivity of 88.503 $\pm$ 3.807%, a specificity of 89.968 $\pm$ 3.287%, a precision of 89.658 $\pm$ 4.078%, and an F1-score of 88.957 $\pm$ 2.350%. These results

TABLE 4. The average classification performances of different pre-trained CNNs with the modified SENet model.

Model	Classification Performance (mean $\pm$ standard deviation)				
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-score (%)
VGG19	88.942 $\pm$ 1.494*	89.459 $\pm$ 2.419*	88.604 $\pm$ 4.123*	88.544 $\pm$ 4.336*	88.890 $\pm$ 1.670*
InceptionV3	78.154 $\pm$ 1.334*	76.688 $\pm$ 4.706*	79.423 $\pm$ 4.524*	78.702 $\pm$ 5.429*	77.509 $\pm$ 3.529*
XceptionNet	80.997 $\pm$ 1.144*	80.043 $\pm$ 1.903*	81.945 $\pm$ 2.637*	81.362 $\pm$ 4.280*	80.616 $\pm$ 2.183*
EfficientNetB0	82.260 $\pm$ 2.662*	78.593 $\pm$ 6.017*	85.444 $\pm$ 7.235*	85.043 $\pm$ 5.862*	81.375 $\pm$ 3.141*
MobileNetV3-Large	92.576$\pm$0.910	92.906$\pm$1.928	92.269$\pm$2.740	92.173$\pm$3.024	92.486$\pm$1.326

*= indicates the difference than the best performer (bold) with statistical significance (p-value < 0.05)

TABLE 5. The average classification performances of the different attention mechanisms.

Model	Average Classification Performance (mean $\pm$ standard deviation)				
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-score (%)
Only MobileNetV3-Large	89.407 $\pm$ 1.458*	88.437 $\pm$ 3.974*	90.471 $\pm$ 4.664	90.276 $\pm$ 4.608*	89.171 $\pm$ 1.820*
MobileNetV3-Large+SENet	90.538 $\pm$ 0.954*	89.926 $\pm$ 2.136*	91.427 $\pm$ 3.330	90.994 $\pm$ 4.228	90.354 $\pm$ 1.432*
MobileNetV3-Large+ECA	89.120 $\pm$ 2.026*	87.522 $\pm$ 5.306*	91.145 $\pm$ 5.285	90.677 $\pm$ 5.519	88.789 $\pm$ 2.332*
MobileNetV3-Large+Modified SENet	92.576$\pm$0.910	92.906$\pm$1.928	92.269$\pm$2.740	92.173$\pm$3.024	92.486$\pm$1.326

*= indicates the difference than the best performer (bold) with statistical significance (p-value < 0.05)

TABLE 6. The comparison of existing deep learning OSA classification models and the proposed model.

Authors	Classifier	Feature/ Representation	Classification Performance (mean $\pm$ standard deviation)				
			Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)
Shen et al. [14]	LSTM	MFCCs	71.435 $\pm$ 4.622*	69.040 $\pm$ 12.019*	73.591 $\pm$ 8.024*	72.390 $\pm$ 7.771*	69.937 $\pm$ 7.780*
Cheng et al. [15]	Four branches of LSTM	MFCCs, Fbanks, Short-time Energy, and LPC	89.171 $\pm$ 1.989*	88.503 $\pm$ 3.807*	89.968 $\pm$ 3.287*	89.658 $\pm$ 4.078*	88.957 $\pm$ 2.350*
Ding et al. [16]	VGG19+LSTM	Mel-Spectrogram	78.879 $\pm$ 2.138*	81.244 $\pm$ 5.707*	76.874 $\pm$ 6.868*	77.923 $\pm$ 6.095*	79.165 $\pm$ 2.444*
The Proposed Method	Pre-trained MobileNetV3-Large +Modified SENet	Mel-Spectrogram and MFCCs	92.576$\pm$0.910	92.906$\pm$1.928	92.269$\pm$2.740	92.173$\pm$3.024	92.486$\pm$1.326

Fbanks = Mel Filter Banks, LPC = Linear Prediction Coefficient, and *= indicates the difference than the best performer (bold) with statistical significance (p-value < 0.05)

highlight the effectiveness of utilizing multiple input features to enhance classification performance. However, the proposed model still achieves the highest accuracy of 92.576 $\pm$ 0.910%, along with highest sensitivity, specificity, precision, and F1-score.

These findings confirm that the proposed approach is not only more effective than existing methods but also more robust and reliable across all evaluation metrics, establishing it as the most suitable model for OSA classification task.

IV. CONCLUSION

This study proposed an attention-based transfer learning model that combines the MobileNetV3 Large with modified SENet. The model utilizes Mel-spectrograms and MFCC features as the inputs to classify snore sounds into normal and abnormal snore events. To assess its effectiveness, the study employed 10-fold cross-validation, using sound segments recorded with an ambient microphone from 70 adult OSA patients of the open-source PSG-Audio dataset [1].

TABLE 7. Patient distribution on 10-fold cross-validation.

	Patient IDs						
Fold1	00001127	00001172	00001234	00001294	00001380	00001388	00001474
Fold2	00001016	00001163	00001193	00001301	00001369	00001398	00001451
Fold3	00001028	00001161	00001232	00001252	00001365	00001396	00001494
Fold4	00001006	00001008	00001097	00001153	00001265	00001290	00001310
Fold5	00001039	00001116	00001131	00001169	00001250	00001340	00001461
Fold6	00001010	00001018	00001143	00001208	00001360	00001428	00001447
Fold7	00001014	00001284	00001316	00001390	00001457	00001463	00001486
Fold8	00001126	00001155	00001224	00001249	00001287	00001331	00001488
Fold9	00001174	00001182	00001217	00001288	00001344	00001392	00001411
Fold10	00001106	00001147	00001178	00001215	00001282	00001342	00001382

= Severe Apnea Patient
 = Moderate Apnea Patient
 = Mild Apnea Patient
 = Normal Patient

The results clearly showed that utilizing both Mel Spectrogram and MFCCs as input features for the proposed model leads to statistically significant improvements in classification performance. The proposed model not only yields the highest accuracy of $92.576 \pm 0.910\%$ but also maintains a strong balance between sensitivity, specificity, precision, F1 score and computational efficiency, making it the strongest candidate for OSA classification task.

Our study is the first to evaluate and compare the previous OSA methods on the same open-source snore dataset. Based on the same data, the proposed method outperformed all the previous models with a smaller number of model parameters. It achieved higher performance metrics across all evaluation criteria. Hence, the proposed model effectively classifies OSA snore sounds with efficiency and robustness.

In summary, the study presents a promising approach to detecting OSA using only audio signals acquired from non-invasive ambient microphones. This approach leverages a resource-efficient, attention-based transfer learning model, which performs strongly in classifying OSA events. The proposed models could potentially enable automatic calculation of the Apnea-Hypopnea Index (AHI) in the future. This method offers a valuable tool for patients to screen themselves at home using portable devices.

Nevertheless, this study employed a dataset comprising only 70 patients. It is advisable to expand the training dataset by incorporating information from a broader patient cohort for future research. Additionally, this model has been trained and evaluated solely on single-sourced datasets, potentially resulting in overfitting and diminished performance when applied to diverse datasets. Consequently, cross-dataset evaluation should be incorporated to assess model generalization across various recording conditions and patient populations.

APPENDIX

Table 7 presents the patient distribution used for 10-fold cross-validation in this study. The dataset is divided into 10 folds, with each fold containing a unique set of 7 patient IDs representing individuals diagnosed with varying levels of OSA severity, i.e. severe, moderate, mild, and normal patients. This diverse distribution ensures that the model is trained and evaluated on different subsets of data from various patients with different OSA conditions across multiple iterations. By doing so, the model is tested on its ability to distinguish between normal and abnormal breathing patterns effectively.

By incorporating patients with varying severity levels into different folds, the model is exposed to a broad range of cases, which enhances its classification accuracy and robustness. The balanced distribution of patients across folds plays a critical role in preventing overfitting, ensuring that the model does not become overly specialized to a specific subset of data. Additionally, this approach provides a fair and comprehensive assessment of the model's classification performance by allowing it to learn from diverse patient groups. The inclusion of severe, moderate, mild, and normal patients further strengthens the model's ability to generalize well across different apnea severity levels.

ACKNOWLEDGMENT

This research was supported by the Petchra Pra Jom Klao Ph.D. Research Scholarship from the King Mongkut's University of Technology Thonburi (Scholarship No. 44/2564).

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